



**SCHOOL OF ENGINEERING AND NATURAL
SCIENCES**

Linear Algebra

Spring 25 - 26

**Project 2
PCA for Data Visualization
Report**

**Under the Supervision of Assoc. Prof. Cem
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STUDENT INFORMATION				GRADE
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- **Objective:**

The aim of this project was to use Principal Component Analysis (PCA) on a dataset describing different countries so we can reduce its high dimensionality and make it easier to visualize and interpret as a 2D matrix. The goal is to highlight the main patterns in the data and show them by only using a few variables while preserving as much information as possible.

- **Introduction/ Explanation:**

In this project, the main problem was dealing with a dataset that has multiple variables, which makes it harder to really understand the relationships and patterns between the countries. To solve this problem, PCA technique was used to reduce that high dimensionality as it transforms the original variables into a new set of uncorrelated variables called principal components, here where data varies the most. From a mathematical point of view, this needs centering the data, computing the covariance matrix, and then finding its eigenvalues and eigenvectors. The eigenvectors of the covariance matrix are the directions of maximum variance in the data, while the eigenvalues say how important each direction is. By projecting the data onto the most significant components, it becomes possible to represent it in a simpler form without losing the main structure.

So what is actually the PCA? PCA shows in which direction the data is spread the most. It is based on transforming the original data into a new coordinate system defined by the eigenvectors of the covariance matrix. This starts by centering the data by subtracting the mean from the data. Centering shifts each feature so its mean = 0, this is required before PCA so no feature dominates just because of its scale of measurement. Then, moving to computing the covariance matrix, which is a 6×6 matrix that measures how much each pair of features varies together:

$$C = [1/(n-1)] X_c^T X_c$$

The principal components are obtained by solving the eigenvalue problem of this matrix after decomposing it, where the eigenvectors represent the directions of maximum variance (the new axes) and the eigenvalues indicate how much variance is taken in each direction, on in other words each axis takes. By selecting the eigenvectors corresponding to the largest eigenvalues after sorting them of course, the data can be projected into a lower-dimensional space while preserving most of its structure. This step only takes the first 2 eigenvectors (columns), forming a 6×2 projection matrix. This is basically the dimensionality reduction step as it takes the very first matrix from 6D down to a new reduced one of 2D without losing its main structure. So when the first eigenvalue is much larger that means that the first principal component captures most variance, so most of the data varies along this direction.

- **Description of the dataset:**

The dataset used in this project consists of 12 fictional countries, each described by six different variables:

- a) GDP
- b) Literacy rate
- c) Life expectancy
- d) Internet usage
- e) CO₂ emissions
- f) Education level

These variables represent different aspects of a country's development on different fields that may be related or not. Since the variables are measured on different scales, the dataset first needed to be centered before using PCA to ensure a fair comparison.

- **Computational steps:**

1. First the dataset was organized into a data matrix X (12x6), where each row represents a country and each column represents a variable (GDP, literacy, life expectancy, internet usage, CO₂ emissions, and education).

$$X = \begin{bmatrix} 55. & 98. & 82. & 92. & 8.1 & 9. \\ 30. & 85. & 74. & 65. & 4. & 6.5 \\ 62. & 99. & 84. & 95. & 9.5 & 9.3 \\ 15. & 70. & 65. & 40. & 2.2 & 4.8 \\ 45. & 92. & 79. & 85. & 6.7 & 8.1 \\ 20. & 75. & 68. & 50. & 3. & 5.5 \\ 58. & 97. & 83. & 90. & 8.9 & 9.1 \\ 25. & 80. & 70. & 60. & 3.8 & 6. \\ 48. & 94. & 80. & 88. & 7.2 & 8.5 \\ 12. & 65. & 60. & 35. & 1.5 & 4.2 \\ 40. & 90. & 77. & 78. & 5.8 & 7.5 \\ 35. & 88. & 75. & 72. & 5. & 7. \end{bmatrix}$$

2. The data was centered by subtracting the mean of each column from the corresponding values, resulting in a centered matrix X_c (12x6). This step ensures that all variables are measured relative to their average. $X_c =$

$$X_c = \begin{bmatrix} 17.91666667 & 11.91666667 & 7.25 & 21.16666667 & 2.625 & 1.875 \\ -7.08333333 & -1.08333333 & -0.75 & -5.83333333 & -1.475 & -0.625 \\ 24.91666667 & 12.91666667 & 9.25 & 24.16666667 & 4.025 & 2.175 \\ -22.08333333 & -16.08333333 & -9.75 & -30.83333333 & -3.275 & -2.325 \\ 7.91666667 & 5.91666667 & 4.25 & 14.16666667 & 1.225 & 0.975 \\ -17.08333333 & -11.08333333 & -6.75 & -20.83333333 & -2.475 & -1.625 \\ 20.91666667 & 10.91666667 & 8.25 & 19.16666667 & 3.425 & 1.975 \\ -12.08333333 & -6.08333333 & -4.75 & -10.83333333 & -1.675 & -1.125 \\ 10.91666667 & 7.91666667 & 5.25 & 17.16666667 & 1.725 & 1.375 \\ -25.08333333 & -21.08333333 & -14.75 & -35.83333333 & -3.975 & -2.925 \\ 2.91666667 & 3.91666667 & 2.25 & 7.16666667 & 0.325 & 0.375 \\ -2.08333333 & 1.91666667 & 0.25 & 1.16666667 & -0.475 & -0.125 \end{bmatrix}$$

3. The covariance matrix was computed to analyze the relationships between variables (6x6):

$$C = [1/(n-1)] X_c^T X_c = \begin{bmatrix} 288.99242424 & 187.17424242 & 126.65909091 & 346.1969697 & 44.94772727 & 29.425 \\ 187.17424242 & 128.08333333 & 85.84090909 & 234.37878788 & 28.875 & 19.49772727 \\ 126.65909091 & 85.84090909 & 58.02272727 & 157.04545455 & 19.61136364 & 13.16136364 \\ 346.1969697 & 234.37878788 & 157.04545455 & 433.42424242 & 53.54090909 & 36.02272727 \\ 44.94772727 & 28.875 & 19.61136364 & 53.54090909 & 7.02386364 & 4.56886364 \\ 29.425 & 19.49772727 & 13.16136364 & 36.02272727 & 4.56886364 & 3.03659091 \end{bmatrix}$$

4. The eigenvalues and eigenvectors of the covariance matrix were calculated. The eigenvectors represent the principal directions of the data, while the eigenvalues define the amount of variance that each direction captures.

Eigenvalues: 6 values

```
[9.08074960x102 8.72332483 1.52388263 2.38533143x10-1 1.72146108x10-2
5.26704979x10-3]
```

Since the first eigenvalue is the largest (908), so that means This means: PC1 alone captures most of the variance in your data and The remaining components explain very little

Eigenvectors: (6x6)

```
[[ 0.5585774 0.80561282 -0.00125256 0.1147447 -0.15931458 -0.02074188]
 [ 0.3733596 -0.31595909 0.59358694 0.62062948 0.15040442 0.02499126]
 [ 0.25111277 -0.09846425 0.59926702 -0.73672277 -0.14392352 -0.06820412]
 [ 0.6889979 -0.46688477 -0.53675135 -0.13008618 -0.02982623 -0.03723536]
 [ 0.08652726 0.15169294 -0.02048091 -0.18176498 0.95763283 -0.13778125]
 [ 0.05771656 0.02169097 0.00324661 -0.09450577 0.11546927 0.98687551]]
```

5. The eigenvalues were sorted in descending order, and the corresponding eigenvectors were rearranged accordingly to identify the most important principal components.
6. The first two principal components were selected, as they capture the majority of the variance in the dataset.

Principal Components (first 2 cols of the Eigenvectors): (6x2)

```
[[ 0.5585774 0.80561282]
 [ 0.3733596 -0.31595909]
 [ 0.25111277 -0.09846425]
 [ 0.6889979 -0.46688477]
 [ 0.08652726 0.15169294]
 [ 0.05771656 0.02169097]]
```

7. The original data was projected onto these two components by multiplying the centered matrix (12x6) with the matrix containing the first 2 columns of the Eigenvectors (6x2), resulting in a lower-dimensional representation of the data (12x2).

Projected Data (2D): (12x2)

```
[[ 31.19675606 0.51132157]
 [-8.73225238 -2.80409637]
 [ 38.18782983 4.45594679]
 [-42.45013702 2.09947329]
 [ 17.62141778 -2.31735791]
 [-30.03751131 -0.27996049]
 [ 31.44723909 4.20294785]
 [-17.88754168 -2.56526822]
 [ 22.4283259 -1.94703334]
 [-50.78841957 3.96997575]
 [ 8.64409627 -2.39792015]
 [ 0.37019704 -2.92802877]]
```

8. Finally, the projected data was visualized using a scatter plot, and a scree plot was created to show how much variance is explained by each principal component.

• **Results:**

Obtained matrices:

a) The given data X (12x6) =

55.	98.	82.	92.	8.1	9.
30.	85.	74.	65.	4.	6.5
62.	99.	84.	95.	9.5	9.3
15.	70.	65.	40.	2.2	4.8
45.	92.	79.	85.	6.7	8.1
20.	75.	68.	50.	3.	5.5
58.	97.	83.	90.	8.9	9.1
25.	80.	70.	60.	3.8	6.
48.	94.	80.	88.	7.2	8.5
12.	65.	60.	35.	1.5	4.2
40.	90.	77.	78.	5.8	7.5
35.	88.	75.	72.	5.	7.

b) the centered data X_c (12x6) =

17.91666667	11.91666667	7.25	21.16666667	2.625	1.875
-7.08333333	-1.08333333	-0.75	-5.83333333	-1.475	-0.625
24.91666667	12.91666667	9.25	24.16666667	4.025	2.175
-22.08333333	-16.08333333	-9.75	-30.83333333	-3.275	-2.325
7.91666667	5.91666667	4.25	14.16666667	1.225	0.975
-17.08333333	-11.08333333	-6.75	-20.83333333	-2.475	-1.625
20.91666667	10.91666667	8.25	19.16666667	3.425	1.975
-12.08333333	-6.08333333	-4.75	-10.83333333	-1.675	-1.125
10.91666667	7.91666667	5.25	17.16666667	1.725	1.375
-25.08333333	-21.08333333	-14.75	-35.83333333	-3.975	-2.925
2.91666667	3.91666667	2.25	7.16666667	0.325	0.375
-2.08333333	1.91666667	0.25	1.16666667	-0.475	-0.125

c) the covariance matrix (6x6) $C = [1/(n-1)] X_c^T X_c =$

288.99242424	187.17424242	126.65909091	346.1969697	44.94772727	29.425
187.17424242	128.08333333	85.84090909	234.37878788	28.875	19.49772727
126.65909091	85.84090909	58.02272727	157.04545455	19.61136364	13.16136364
346.1969697	234.37878788	157.04545455	433.42424242	53.54090909	36.02272727
44.94772727	28.875	19.61136364	53.54090909	7.02386364	4.56886364
29.425	19.49772727	13.16136364	36.02272727	4.56886364	3.03659091

d) Eigenvalues: 6 values

9.08074960×10^2	8.72332483	1.52388263	$2.38533143 \times 10^{-1}$	$1.72146108 \times 10^{-2}$	$5.26704979 \times 10^{-3}$
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e) Eigenvectors: (6x6)

```
[[ 0.5585774  0.80561282 -0.00125256  0.1147447 -0.15931458 -0.02074188]
 [ 0.3733596 -0.31595909  0.59358694  0.62062948  0.15040442  0.02499126]
 [ 0.25111277 -0.09846425  0.59926702 -0.73672277 -0.14392352 -0.06820412]
 [ 0.6889979 -0.46688477 -0.53675135 -0.13008618 -0.02982623 -0.03723536]
 [ 0.08652726  0.15169294 -0.02048091 -0.18176498  0.95763283 -0.13778125]
 [ 0.05771656  0.02169097  0.00324661 -0.09450577  0.11546927  0.98687551]]
```

g) Principal Components (first 2 cols of the Eigenvectors): (6x2)

```
[[ 0.5585774  0.80561282]
 [ 0.3733596 -0.31595909]
 [ 0.25111277 -0.09846425]
 [ 0.6889979 -0.46688477]
 [ 0.08652726  0.15169294]
 [ 0.05771656  0.02169097]]
```

h) Projected Data (2D): (12x2)

```
[[ 31.19675606  0.51132157]
 [-8.73225238 -2.80409637]
 [ 38.18782983  4.45594679]
 [-42.45013702  2.09947329]
 [ 17.62141778 -2.31735791]
 [-30.03751131 -0.27996049]
 [ 31.44723909  4.20294785]
 [-17.88754168 -2.56526822]
 [ 22.4283259 -1.94703334]
 [-50.78841957  3.96997575]
 [ 8.64409627 -2.39792015]
 [ 0.37019704 -2.92802877]]
```

i)

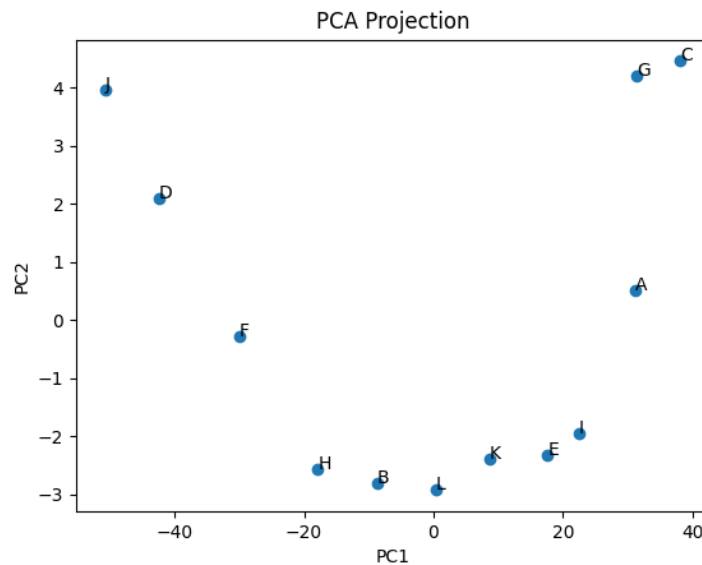


Figure 1: PCA Projection Plot that shows all 12 countries plotted in 2D space after reducing from 6 dimensions.

Notice the letters follow alphabetical order matching the countries as they appear in the dataset table, for example A → Alandia, B → Borland, etc.

X-axis (PC1) → the most important pattern in the data (captures about 98% of variance since eigenvalue was 908).

Y-axis (PC2) → the second pattern (captures much less variance)

So here the right Vs. left position matters the most

Far right → High values overall (rich, educated, healthy countries)

Far left → Low values overall (poorer, less developed countries)

Which means Countries close together have similar socioeconomic profiles, while Countries far apart have very different profiles.

Table 1: PCA projection plot explanation.

Group	Countries	Why
High development	C, A, G	Cluster on the right
Mid development	E, I, K	Sit in the middle
Low development	D, J, F	Cluster on the left

Even though 6 different things were measured about each country, they all really just vary along one main axis, development level.

j)

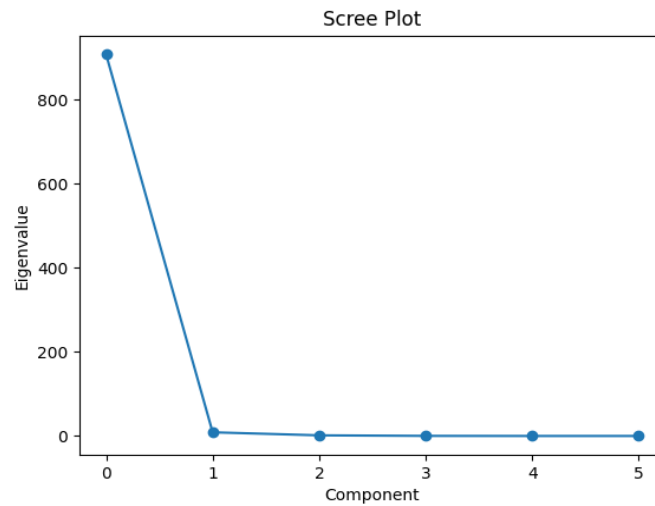


Figure 2: The Scree Plot shows how much variance each principal component captures, so it helps to decide how many components are worth keeping.

X-axis → the component number (from 0 to 5, for the 6 components)

Y-axis → the eigenvalue is how much variance that component explains

A massive drop was noticed after the first component (from 0 to 1), then nearly flat.

Table 2: Scree plot explanation.

Point	Eigenvalue	Meaning
0	908	PC1 explains almost everything
1	8.72	PC2 adds a tiny bit
2	1.52	PC3 almost negligible
3-5	≈ 0	Basically useless

And by following the elbow rule to look for where the curve bends like an elbow and goes flat and that will be the cutoff point. In here it will be after 0 there is a massive drop so the clear elbow is seen at 1. This means that 1 component is enough to represent the data.

So, All 6 features of the countries can be summarized by just a single number, a development score.

- Results interpretation:**

Starting with the centered data, subtracting the mean shifted all variables so they are measured around zero. This step was important as it removed any bias caused by different scales, making all variables contribute fairly to the analysis.

Looking at the covariance matrix, most of the values are positive, which indicates that the variables are positively correlated. This means that when one variable increases (like GDP),

others such as education, literacy, and life expectancy also tend to increase. This already tells that there is a hidden common factor connecting these variables.

Moving to the eigenvalues, there is a very large first eigenvalue (around 908), while the remaining ones are very small, relatively. This shows that the first principal component captures almost all of the variance in the dataset and obviously the one in control, while the rest contribute very little. In other words, most of the information in the data can be explained by a single main direction.

The eigenvectors represent the directions of these principal components. From the first eigenvector, we can see that most variables contribute positively, which means that this component is combining all the variables into one general measure. This is why it can be interpreted as an overall “development level” of each country.

After selecting the first two principal components, the data was projected into a 2D space. From the projected data values, we can notice a clear differentiation between countries based on their scores along the first component. Countries with large positive values are those with higher development indicators, while those with large negative values have lower indicators.

This becomes clearer in the PCA projection plot, where most of the variation is along the horizontal axis (PC1), since it explains about 98% of the variance. This means that the horizontal direction is the most important, and it mainly represents the overall development level of the countries. Countries are spread from left to right, with high-development countries on the right and low-development ones are on the left. Countries on the right side of the plot have higher values in terms of GDP, education, literacy, and life expectancy, while countries on the left side have lower values. This shows that PC1 can be interpreted as a general “development axis” that differentiates more developed countries from less developed ones. However, similarity between countries is not determined only by being on the same side. It depends on how close the points are in both PC1 and PC2. For example, countries C and G are very close to each other, which indicates that they can be similar in their overall characteristics, but C is better as it more on the right. Similarly, E and K are also relatively close while E is better, and B and L appear close as well and L is better, showing similar patterns.

On the other hand, some countries that are on the same side are not actually that similar. For example, A is on the right side like C and G, but it is clearly away from them along the vertical axis (PC2), meaning it differs in some secondary characteristics. The same idea applies on the left side, where countries like D, J, and F are not that grouped and show noticeable differences between them.

Looking at the second principal component (PC2), it captures much less variance, but it still helps distinguish between countries that have similar overall development but differ in specific aspects. This is why some points that are close horizontally are still separated vertically.

Finally, the scree plot shows a sharp drop after the first component and then becomes nearly flat. This means that the first component alone explains almost all the variance, and adding more components does not improve the representation that much. Based on this, it is clear that one or two components are enough to describe the dataset.

Overall, even though the data originally had six variables, the results show that most of the structure can be explained by a single dominant pattern, while the second component only adds small refinements and details to distinguish between similar countries.

- **Reflection/ Discussion:**

Overall, PCA was very useful in simplifying the dataset and making it much easier to understand. Even though the data originally had six different variables (6D), PCA showed that most of the variation can actually be explained by just one main component. So, it reduces the complexity of the data without completely losing its meaning.

In this case, PCA mainly preserved the general patterns and relationships between the countries. For example, countries that are close to each other in the plot, like C and G or E and K, still appear similar after the transformation. At the same time, countries that are far apart clearly show differences in their characteristics.

However, some information was lost in the process, especially the smaller variations captured by the other lower principal components (PC2, PC3, etc.). In addition to small differences between countries that are very similar, they might look almost the same after reduction. Also some detailed relationships between variables that are not part of the main pattern. Besides, the exact contribution of each original variable becomes less clear after transformation. And any info that does not align with the main direction of the variance PC1 as well. These details are not completely removed, but they become less important when we reduce the dimensions. Also, even though PC2 helps separate some points vertically, it still captures much less information compared to PC1, so some details are not fully represented.

Looking at the first principal component (PC1), it is clear that variables like GDP, internet usage, literacy, and education dominate it as they all contribute positively and strongly. This is the reason why PC1 can be interpreted as a general “development level” for the countries.

From the PCA plot, countries are mainly differentiated along the first principal component, which represents the overall development level. However, not all countries on the same side are exactly very similar. For example, countries like C and G are very close to each other, which indicates strong similarity, while others like A are on the same side but still relatively far apart. This shows that similarity depends on both principal components, not just one. Therefore, PCA helps identify general patterns, but smaller differences between countries can still be seen in the second component.

In terms of variance, the first principal component alone explains almost all of the variance (around 98%), while the second component explains only a very small amount. This means that using just the first two principal components is already enough to represent the dataset very well, and even using only one component could still give a good overall idea.

To sum up, PCA makes it much easier to visualize and interpret complex data by reducing dimensions, but it always comes with a trade-off between simplicity and losing some less important details.

Table 3: comparing PC1 of the variables.

Variable	PC1 Loading	Dominance
Internet	0.689	Strongest
GDP	0.559	Very strong
Literacy	0.373	Moderate
Life Expectancy	0.251	Moderate
CO ₂	0.087	Weak
Education	0.058	Weakest

Internet and GDP dominate PC1 the most, followed by Literacy and Life Expectancy. All loadings are positive, meaning PC1 is essentially a combined "overall development score", so the countries that are high in one tend expected be high in all of them. This is why PC1 can be read as a single development axis.

Table 4: From the projected 2D data, countries that are close in both PC1 and PC2 coordinates are similar

Similar Pair	PC1 Scores	Why Similar
C & G	38.2 vs 31.4	Closest high-development pair
A & G	31.2 vs 31.4	Nearly identical PC1
E & I	17.6 vs 22.4	Both mid-development
B & L	-8.7 vs 0.37	Similar low-mid range
D & J	-42.5 vs -50.8	Both low-development

Note that A (Alandia) differs from C (Cyperia) and G (Galvia) vertically (PC2), meaning A shares the same overall development level but differs in some secondary characteristic. Similarly, D (Deltora), J (Jovinia), and F (Freonia) are all on the low side but are not tightly grouped.

Table 5: comparing using the eigenvalues

Component	Eigenvalue	Variance Explained
PC1	908.07	98.07%
PC2	8.72	0.94%
PC3-PC6	~1.75 as total	~0.19%
PC1 + PC2 combined	—	≈ 99.01%

Both the first two PCs explain about 99% of all variance. This is exceptionally high, meaning the 2D scatter plot produced is an almost perfect representation of the original 6D data. Almost nothing is lost in the reduction (1%) which also confirms that all 6 variables are so strongly correlated that one number (development level) explains nearly everything.